

Study Drugs Among University Students

1 Main article about the societal challenge

The news article “Campus researchers address student usage of ‘study drugs’ and potential consequences” was written by Rachel Goldsmith and published on 19 April 2026 in *Pipe Dream*, Binghamton University’s independent student newspaper. The article is in English, so no translation is required.

[Read the original Pipe Dream article](#)

2 The challenge identified in the news article

The abuse of prescription “performance” drugs like Adderall or Medikinet by students has become a concern on university campuses in many Western countries. The article presents the issue on many levels, including the reasons and risks, as well as a more personal approach with personal accounts.

The article’s informal survey describes its respondents rather than establishing prevalence among all students. Appropriate prescribed treatment should be distinguished from non-medical use.

2.1 Why are we interested in this topic?

As students, we encounter pressure to perform at the highest level almost every day. It is not uncommon to hear that someone we know has tried these types of solutions. This is worrying in two ways: on one hand, we worry about close ones dabbling in substances; on the other, we fear that we will be left behind academically, creating a kind of snowball effect.

2.2 For whom is this challenge relevant?

This challenge is relevant to a wide range of people: students, professors and university staff, parents, and employers. Doctors and healthcare providers are also relevant stakeholders because they prescribe medicines and support students who may experience medication-related problems.

3 Selected articles and their relevance

3.1 Article 1: Evidence for cognitive enhancement

The scientific article “Prescription stimulants’ effects on healthy inhibitory control, working memory, and episodic memory: A meta-analysis was written by Irena P. Ilieva, Cayce J. Hook, and Martha J. Farah and published in 2015 in *Journal of Cognitive Neuroscience*.

This article is relevant because it tests the belief that prescription stimulants improve the cognitive abilities used in studying. The authors analyse 48 studies involving 1,409 participants and report benefits on some tasks, while concluding that overall enhancement is probably modest. Publication-bias concerns affect some findings. This helps us distinguish perceived productivity

from demonstrated cognitive improvement. Laboratory performance does not automatically establish better grades or long-term benefits.

- **Title:** Prescription stimulants’ effects on healthy inhibitory control, working memory, and episodic memory: A meta-analysis.
- **Authors:** Irena P. Ilieva, Cayce J. Hook, and Martha J. Farah.
- **Year:** 2015.
- **Journal:** Journal of Cognitive Neuroscience.
- **Type:** Peer-reviewed meta-analysis.
- **DOI:** [10.1162/jocn_a_00776](https://doi.org/10.1162/jocn_a_00776).

3.2 Article 2: Prevalence of study-drug use

This article is highly relevant to our societal challenge because it provides scientific evidence about the use of study drugs among university students. It shows that although the reported prevalence is relatively low, the problem is difficult to measure accurately because the available studies do not use representative sampling. Students might also not feel comfortable admitting non-medical drug use. The article highlights the difference between the way the media portrays the problem and the scientific evidence.

- **Title:** Prevalence of the use of prescription stimulants as “study drugs” by UK university students: A brief report.
- **Authors:** Ffinian Jones and Philip M. Newton.
- **Journal:** Brain and Behavior.
- **Year:** 2024.
- **Type:** Peer-reviewed brief report reviewing survey studies.
- **DOI:** [10.1002/brb3.3419](https://doi.org/10.1002/brb3.3419).

[Read the full article on PubMed Central](#)

3.3 Article 3: Benefits and harms of smart drugs

The scientific article “Benefits and harms of ‘smart drugs’ (nootropics) in healthy individuals” was written by Fabrizio Schifano and colleagues and published in 2022 in *Drugs*, a peer-reviewed medical journal. It is a review article.

The paper describes how cognitive enhancers, including prescription stimulants like amphetamines and methylphenidate, affect healthy people, especially students taking them as study drugs. The authors explain that evidence of enhancement in healthy individuals remains uncertain, while these drugs can carry serious harms. By influencing neurotransmitter systems, their use may be associated with cardiovascular, neurological and psychological complications, addictive behaviour, and adverse cognitive effects. This article helps us understand the situation by showing the potential pharmacological risks students take when turning to stimulants for academic success.

- **Title:** Benefits and harms of ‘smart drugs’ (nootropics) in healthy individuals.
- **Authors:** Fabrizio Schifano, Valeria Catalani, Safia Sharif, Flavia Napoletano, John Martin Corkery, Davide Arillotta, Suzanne Fergus, Alessandro Vento, and Amira Guirguis.
- **Journal:** Drugs.

- **Year:** 2022.
- **Type:** Peer-reviewed review article.
- **DOI:** [10.1007/s40265-022-01701-7](https://doi.org/10.1007/s40265-022-01701-7).

3.4 Article 4: Social-media markets for prescription drugs

This article is relevant to our societal challenge because it examines prescription-drug distribution through social-media markets. It provides a perspective on how access to these drugs can occur outside formal healthcare settings. This helps us understand why prescription records alone cannot capture all non-medical use. The study concerns Swedish Facebook markets, so its findings should not be treated as a prevalence estimate for university students.

- **Title:** Social media markets for prescription drugs: Platforms as virtual mortars for drug types and dealers.
- **Authors:** Jakob Demant, Silje Anderdal Bakken, and Alexandra Hall.
- **Journal:** Drugs and Alcohol Today.
- **Year:** 2020.
- **Type:** Peer-reviewed original research article.
- **DOI:** [10.1108/DAT-06-2019-0026](https://doi.org/10.1108/DAT-06-2019-0026).

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References

- Goldsmith, R. (2026, April 19). Campus researchers address student usage of ‘study drugs’ and potential consequences. *Pipe Dream*. [Read the original Pipe Dream article](#)
- Ilieva, I. P., Hook, C. J., & Farah, M. J. (2015). Prescription stimulants’ effects on healthy inhibitory control, working memory, and episodic memory: A meta-analysis. *Journal of Cognitive Neuroscience*, 27(6), 1069–1089. doi: [10.1162/jocn_a_00776](https://doi.org/10.1162/jocn_a_00776)
- Jones, F., & Newton, P. M. (2024). Prevalence of the use of prescription stimulants as “study drugs” by UK university students: A brief report. *Brain and Behavior*, 14(2), e3419. doi: [10.1002/brb3.3419](https://doi.org/10.1002/brb3.3419)
- Schifano, F., Catalani, V., Sharif, S., Napoletano, F., Corkery, J. M., Arillotta, D., Fergus, S., Vento, A., & Guirguis, A. (2022). Benefits and harms of ‘smart drugs’ (nootropics) in healthy individuals. *Drugs*, 82(6), 633–647. doi: [10.1007/s40265-022-01701-7](https://doi.org/10.1007/s40265-022-01701-7)
- Demant, J., Bakken, S., & Hall, A. (2020). Social media markets for prescription drugs: Platforms as virtual mortars for drug types and dealers. *Drugs and Alcohol Today*, 20(1), 36–49. doi: [10.1108/DAT-06-2019-0026](https://doi.org/10.1108/DAT-06-2019-0026)

A Original newspaper article

The following screenshots preserve the original English article supplied with the report. The two views overlap slightly to keep text at the page boundary readable.



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NEWS

Campus researchers address student usage of 'study drugs' and potential consequences



Researchers at Binghamton University have found that stressed students who use Adderall tend to believe it is a safer alternative to caffeine, cannabis and cocaine, but consequences can create serious short-term and long-term challenges.

By RACHEL GOLDSMITH | APRIL 19, 2024

The use of "study drugs" to promote focus while tackling papers and exams has become ubiquitous on college campuses. Adderall, a central nervous stimulant prescribed to treat attention-deficit/hyperactivity disorder or narcolepsy, is commonly abused by students to increase productivity and academic performance.

In response to Pipe Dream's informal drug survey, 28 out of 78, or approximately 36 percent of responders, said they have used Adderall, Ritalin or another stimulant before studying.

"I took Adderall at 10 p.m. the night before my chem final and could not sleep the entire night before it," said one anonymous responder.

Binghamton Student Managed Adderall Research Team is a student-run research group focused on the abuse of study drugs and was founded in 2018 by Lina Begdache Ph.D. '08, an associate professor of health and wellness studies. Research from the group found that students who use Adderall as a study drug do so to promote concentration and determined that BU students often think Adderall is safer than caffeine, marijuana and cocaine because it is a prescription drug, Begdache told Pipe Dream.

According to a 2024 study led by Begdache, using study drugs can lead to addiction to cannabis, alcohol and other substances.

"Substance use promotes the release of the neurotransmitter dopamine, which is responsible for the initial euphoria and feelings of pleasure," Begdache told BingUNews. "These sensations act as a positive reinforcement for further substance use. The continuous activation of the limbic system through drug use leads to dependence, in a sense that this substance is no longer producing pleasurable feelings. Individuals have to either increase the dose or resort to something more potent."

B-SMART researcher Zachary Costanza, a sophomore majoring in integrative neuroscience, explained that college students are particularly susceptible to using study drugs.

"Ultimately, college is really the perfect storm for many of the risk factors that are associated with Adderall use," Costanza wrote in a statement to Pipe Dream. "Students, especially here at Binghamton, face intense academic pressures, sleep deprivation, and an environment that often glorifies productivity at all costs. These factors, combined with the misconceptions that Adderall will improve academic outcomes, are exactly why stimulants get framed socially as an effective shortcut."

Costanza described the various negative effects of adderall abuse for young people. Adderall increases dopamine levels in the brain, which "mimic[s] the effects of many other recreational drugs" and can increase the likelihood of addiction. Costanza also noted that "this can cause students to feel less

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4

Original newspaper article (continued)

face intense academic pressures, sleep deprivation, and an environment that often glorifies productivity at all costs. These factors, combined with the misconceptions that Adderall will improve academic outcomes, are exactly why stimulants get framed socially as an effective shortcut."

Costanza described the various negative effects of adderall abuse for young people. Adderall increases dopamine levels in the brain, which "mimic[s] the effects of many other recreational drugs" and can eventually lead to a decrease in natural dopamine production. This can cause people to feel less motivation or focus overall, according to Costanza.

Chris Downey, the director of the University Counseling Center, explained some of the reasons University students might use stimulants.

"These may include academic pressure, attempts to improve efficiency, difficulty balancing energy and social/academic demands, self-managing underlying mental health concerns, concentration problems, euphoria or thrill-seeking, and cultural normalization," Downey wrote in an email to Pipe Dream. "Study drugs can seem appealing as the impacts can be fast-acting and powerful, but their consequences can create short-term or long-term challenges and are rarely sustainable."

The UCC provides resources for students struggling with study drugs, including a space to discuss the issue, identify healthier alternatives and perform a consultation for ADHD or significant attention problems. For students seeking treatment, the UCC offers individual counseling with an addiction specialist, group therapy and referrals to off-campus addiction service providers.

Begdache emphasized the dangers of reliance on Adderall as a study drug, arguing that it disrupts sleep, affects long-term memory and can permanently change brain function in young people. She recommends those abusing Adderall to "seek help right away" to prevent irreversible damage.

She also explained that because of the drug's impact on long-term memory, its use might not result in improved academic performance.

"People take Adderall to improve their grades and we found that those who use it on a regular basis tend to have lower GPAs," Begdache said.

While abusing study drugs can lead to dependency and other detriments to students, stimulants are often an effective treatment for certain conditions. However, even prescribed Adderall might have negative effects, Begdache added, and as a dietitian, she recommends physicians consider lifestyle factors before prescribing.

J. David Jentsch, a distinguished professor and chair of psychology and researcher of addictive substances, said that Adderall will have negative effects on anyone taking it, but in some instances, like for patients with ADHD, its positive impacts will outweigh the negative consequences. Dosage adjustments by medical professionals can also limit negative effects, he added.

"Any kind of prescription medication, there's a reason they're prescribed, because we need advice and consent in order to use them," said Jentsch.

Editor's Note (4/22): A previous version of this article listed Lina Begdache as an assistant professor of health and wellness studies. Begdache is an associate professor.

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